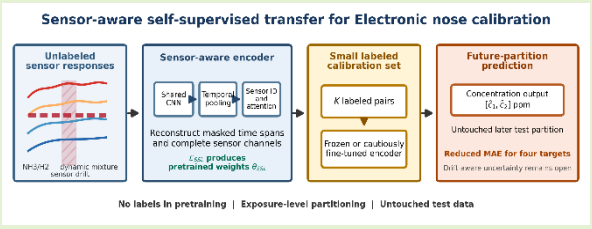
# Sensor-Aware Self-Supervised Transfer Learning for Label-Efficient Multigas Electronic Nose Calibration

Dinh-Van Nguyen, DangThiThu Ha, Thu-Huong Nguyen , and NguyenDuc Hoa

***Abstract*— Electronic nose calibration usually relies on controlled gas exposures with known concentrations. In practice, however, unlabeled responses are much easier to collect because they carry no concentration annotation. The present experiments reuse controlled exposures with their concentration labels withheld during pretraining, so they demonstrate reduced labeled-calibration requirements rather than reduced gas-generation cost. This study explores whether those unlabeled measurements can be useful before the calibration stage through sensor-aware self-supervised learning. The proposed model uses a shared dilated one-dimensional convolutional encoder to learn temporal behavior. Sensor embeddings preserve channel identity, while attention captures relationships across the array. During pretraining, the model reconstructs masked time spans and complete sensor channels. The resulting encoder is then adapted using a limited number of concentration labels. The evaluation includes a private 20-channel NH3/H2 array, a public 16-channel CO/ethylene array, and a public 36-month sensor drift dataset. At the full label budget, transfer achieves mean absolute errors of 14.92 ppm for NH3, 29.16 ppm for H2, 46.16 ppm for CO, and 1.371 ppm for ethylene. These results correspond to gains of 19.2%, 20.3%, 29.0%, and 31.2% over scratch training. The benefit generally becomes larger when fewer labels are available. On sensor drift dataset, oracle-class gas-conditional concentration error decreases in three future partitions by 17.3%, 29.9%, and 2.8%. Quantitative transfer therefore retains some within-gas concentration information under long-term drift, although the benefit decreases under the strongest future shift.**




***Index Terms*— Electronic nose, self-supervised learning, transfer learning, gas concentration regression, sensor drift, label-efficient calibration, sensor attention.**

## I. Introduction

Electronic nose systems infer gas properties from the combined response of partially selective sensors. For concentration estimation, they are normally calibrated through controlled exposures with known compositions. Collecting these labels becomes costly as the number of gases, mixtures, and measurement sessions grows. Moreover, aging sensors may require the calibration process to be repeated.

Meanwhile, unlabeled measurements are often available in much larger quantities. Although they do not provide direct concentration targets, they still contain temporal response shapes and recurring relationships among sensors. It is therefore reasonable to ask whether these patterns can be learned first and then reused when a smaller labeled calibration set becomes available.

Several earlier studies have already made use of limited or unlabeled Electronic nose data. M-Training [1] and M-S4VMs [2], for example, use pseudo-labeling for gas recognition. Drift research has taken other directions, including domain adaptation, few-shot learning, and meta-learning [3]–[7]. Quantitative calibration appears less often in this literature, even though it is important for monitoring applications.

There is also evidence that temporal responses contain useful concentration information. Reservoir computing has tracked changing concentrations despite slow sensor dynamics [8]. More recently, CNN-LSTM models and lightweight transferred networks have supported both recognition and concentration estimation [9], [10]. Small-data transfer and domain-adaptive regression offer further options when calibration data are limited [11], [12].

In this work, the sensor array is treated as more than an anonymous multivariate series. Each channel belongs to a physical sensor and may have its own response behavior. Accordingly, a shared temporal encoder learns common dynamics, while sensor identities and cross-sensor attention preserve the organization of the array. Pretraining hides time spans and complete channels, which must be recovered from the remaining measurements. The encoder is later frozen or cautiously fine-tuned for concentration regression. The study tests this approach on two raw-response datasets and across several label budgets. It also follows the public sensor drift dataset into future partitions. Transfer improves two-gas

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regression and oracle-class gas-conditional concentration estimates under drift. The complete pipeline is shown in Fig. 1.

## II. Related Work

Quantitative estimation becomes important when an Electronic nose is used as a measuring instrument rather than only as a recognizer. Fonollosa et al. [8] used reservoir computing to estimate rapidly changing concentrations from slower sensor responses. Their study drew attention to transient dynamics, which may be lost when a response is reduced to a steady-state value.

Deep models have since combined recognition and regression in several ways. Ni et al. [9] used GAF-MTF representations with a CNN-LSTM model, reporting 95.31% classification accuracy and an R^2 of 0.9510. Tang et al. [10] instead paired a single MEMS sensor with a lightweight transferred network. Work on smaller datasets includes parameter-feature transfer [11], while OT-DARN uses teacher-student domain adaptation with selected pseudo-labels [12].

M-Training [1] and M-S4VMs [2] bring unlabeled samples into Electronic nose recognition through predicted labels. This strategy can be helpful when labels are scarce, although incorrect predictions may be reinforced as training continues.

Self-supervised learning avoids that particular dependence by taking its pretraining target from the measurements themselves. Common time-series objectives include masking, reconstruction, contrastive learning, and adversarial learning [13]. Masked reconstruction is a natural fit for sensor arrays because missing intervals require temporal context. Likewise, missing channels encourage the model to use information shared across sensors. CEGNN [6] takes a different route by combining contrastive learning, graph reasoning, and a small reference set.

In addition to accurately estimating gas concentrations, sensor drift due to aging is also a challenging problem. Sensor drift gradually changes the response distribution, so models trained at one time may not behave in the same way later. TDACNN [3] addresses source-target differences, whereas SAD-CNN [5] combines a one-dimensional CNN with adversarial adaptation and pseudo-labeling. MDCN [4] uses metric learning with a small reference set. Other studies extend the setting through contrastive graph reasoning [6] or model-agnostic meta-learning [7], including tests involving class asymmetry, different devices, and future drift. Most evaluations on the 36-month benchmark emphasize whether gas identity survives the shift, while concentration behavior receives less attention. For this reason, we use its future partitions to study oracle-class gas-conditional concentration regression under temporal drift.

Overall, these studies leave unlabeled representation learning and quantitative calibration only partly connected. The present study bridges that gap by first learning sensor-specific temporal and cross-channel structure without concentration labels, and then reusing the encoder with a small labeled calibration set. The future-partition evaluation further examines whether this quantitative transfer remains useful as the sensors drift, and where further adaptation is needed. The proposed approach is expected to reduce the labeled-calibration effort for electronic nose systems, and the drift evaluation tests how far the self-supervised representation stays useful as the sensors age.

## III. Proposed Method

### A. Problem Formulation

For the two raw-response datasets, one gas exposure is represented as the multichannel time series in (1), with the two-gas concentration target in (2)

$$X_i \in \mathbb{R}^{T\times S}, \qquad (1)$$

where $T$ is the number of temporal samples and $S$ is the number of physical sensors. The target is a two-dimensional concentration vector,

$$y_i = \begin{bmatrix} y_{i1} \\ y_{i2} \end{bmatrix} \in \mathbb{R}^2. \qquad (2)$$

For the NH3/H2 dataset, $T = 64$ and $S = 20$. For the dynamic gas mixture dataset, $T = 64$ and $S = 16$.

The goal is not simply to train a regressor on all labeled data. Instead, we assume that a larger set of unlabeled exposures is available before the final calibration stage. A small adaptation set is written as in (3), and the pretrained encoder and regression head are then adapted under the objective in (4)

$$\mathcal{A}_K = \{(X_i, y_i)\}_{i=1}^{K}, \qquad (3)$$

where $K$ controls the labeled calibration budget. After pretraining, the encoder $f_\theta$ and regression head $g_\phi$ are optimized by

$$(\theta^*, \phi^*) = \underset{\theta,\phi}{\operatorname{argmin}} \frac{\sum_{(X_i,y_i)\in\mathcal{A}_K} \ell_{\text{Huber}}\left(g_\phi(f_\theta(X_i)), y_i\right)}{|\mathcal{A}_K|} \qquad (4)$$

Depending on the adaptation policy, $\theta$ is either randomly initialized, frozen after self-supervised pretraining, or updated with a lower learning rate.

### B. Shared Temporal Sensor Encoder

The first stage extracts a temporal representation for each sensor. We use the same dilated 1-D convolutional encoder $\psi_\theta$ for all physical channels. For sensor $s$ in exposure $i$,

$$H_{is} = \psi_\theta(X_i[:, s]). \qquad (5)$$

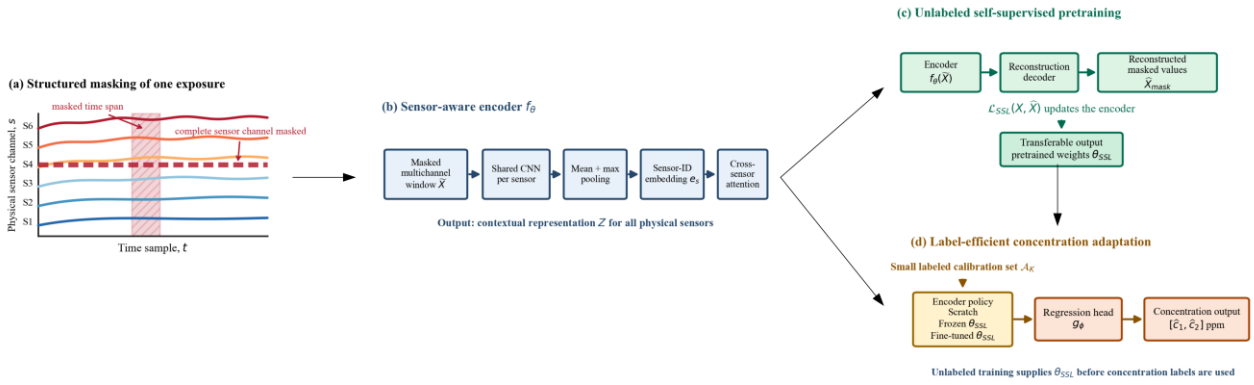



Fig.1. Full sensor-aware pretraining, adaptation pipeline.

The per-sensor temporal encoding in (5) is summarized by combining mean and maximum pooling into the sensor token in (6),

$$h_{is} = [\mathrm{MeanPool}(H_{is}); \mathrm{MaxPool}(H_{is})]. \quad (6)$$

Mean pooling captures the overall response level, while maximum pooling keeps stronger local activation. The shared encoder also limits the number of sensor-specific parameters. Thanks to shared weights, the model can learn common response dynamics from all channels rather than fitting a separate temporal model for each sensor.

The implemented temporal encoder receives two channels per sensor: the scaled signal and its binary observation mask. It applies three convolutions with channel widths $2 \to 24 \to 32 \to 32$, kernel sizes (5,5,3), and dilations (1,2,4), respectively. Padding preserves the 64-sample length, and GELU follows each convolution. Mean and maximum pooling produce 64 features, which are projected through a GELU-activated linear layer to a 32-dimensional sensor token.

### C. Physical Sensor Identity Encoding

Sharing the temporal network introduces a possible ambiguity. Two different physical sensors may produce temporal features with a similar numerical pattern, even though their selectivity is different. We therefore attach a learned identity embedding $e_s$ to each sensor token,

$$z_{is} = h_{is} + e_s. \quad (7)$$

The identity embedding in (7) is learned jointly with the encoder. It does not contain hand-coded gas knowledge. Instead, it gives the model a stable way to know which physical channel generated a token. This is a useful inductive bias for heterogeneous sensor arrays because channel identity is part of the sensing system itself.

The sensor-ID mechanism is deliberately simple. It does not assume that one sensor is selective to one gas. It only preserves channel meaning and lets downstream attention learn how each sensor participates in the array response.

### D. Cross-sensor Self-Attention

The sensor tokens are stacked as in (8)

$$Z_i = [z_{i1}, z_{i2}, \dots, z_{iS}]^\top \in \mathbb{R}^{S\times d}. \quad (8)$$

We then apply Transformer-style self-attention across the sensor dimension. Query, key, and value matrices are given by (9),

$$Q = Z_i W_Q, \quad K = Z_i W_K, \quad V = Z_i W_V, \quad (9)$$

and the attention output is given by (10)

$$A_i = \mathrm{softmax}\left(\frac{QK^\top}{\sqrt{d_k}}\right)V. \quad (10)$$

This layer allows one sensor token to be interpreted in the context of the rest of the array. The motivation is physical. A gas mixture is not identified from one sensor in isolation; it is inferred from a pattern of cross-sensitive responses. Attention gives a flexible mechanism for learning this pattern.

The implementation uses two pre-normalized Transformer encoder layers. Each layer has a 32-dimensional token size, four attention heads, a 96-unit feed-forward sublayer, GELU activation, and dropout 0.1. A final layer normalization is applied to the contextual sensor tokens. For raw-response regression, the (S) token matrix is flattened; it is not pooled across sensors. The regression head has dimensions (32S), GELU activations, and dropout 0.15 after the 128-unit layer.

### E. Structured Masked Self-Supervision

The self-supervised task hides physically meaningful parts of the input and asks the model to reconstruct them. For raw time-series data, we use two masking modes.

First, contiguous temporal spans are removed. For mask $M$,

$$M_{bts} = 0 \quad \text{if} \quad a_k \le t < a_k + w_k \quad \text{and} \quad Z_{bks} = 1, \quad (11)$$

The temporal-span mask in (11) uses a span length sampled as in (12),

$$w_k \sim \mathcal{U}\{6, \dots, 16\}, \quad (12)$$

and per-sensor participation follows (13)

$$Z_{bks} \sim \mathrm{Bernoulli}(0.45). \quad (13)$$

Three temporal spans are sampled for each example. This masking forces the encoder to infer missing local behavior from the remaining response.

Second, one to three full sensor channels are hidden. No temporal value from those sensors remains visible, so reconstruction must draw on the rest of the array. Besides encouraging cross-sensor learning, this setup offers an initial test of how the representation behaves when channel information is absent.

The masked reconstruction loss is given by (14):

$$\mathcal{L}_{\mathrm{SSL}} = \frac{\sum(1-M)(\hat{X}-X)^2}{\max(1, \sum(1-M))} + 0.05\frac{\sum M(\hat{X}-X)^2}{\sum M}. \quad (14)$$

The first term emphasizes the hidden entries. The second provides a smaller consistency penalty on visible values. We also concatenate the mask with the corrupted signal, as in (15),

$$X^{\mathrm{in}} = [X \odot M,\ M], \quad (15)$$

so that a masked value is not confused with a real zero. This avoids need for target pseudo-labels during pretraining.

### F. Adaptation Policies and Calibration-Pair Selection

To integrate pretrain result with labeled data, three adaptation policies are selected: scratch, frozen transfer and fine-tuned transfer. Scratch trains the whole network from random initialization using only labeled calibration data. Frozen transfer loads the pretrained encoder and updates only the supervised prediction head. Fine-tuned transfer initializes from the pretrained encoder and updates it with a lower learning rate while the prediction head is trained normally. Calibration pairs are selected by the farthest-point rule in (16).

The calibration pairs are selected to cover the available concentration space by farthest-point sampling, following the limited-experiment calibration motivation in [14]. Let $\mathcal{P}$ be the candidate set and $\mathcal{S}_k$ the selected set after $k$ steps. The next pair is

$$p_{k+1} = \arg\max_{p\in\mathcal{P}\setminus\mathcal{S}_k} \min_{q\in\mathcal{S}_k} \| \tilde{p} - \tilde{q} \|_2, \quad (16)$$

where $\tilde{p}$ denotes the normalized concentration pair.

## IV. Datasets and Experiment Protocol

### A. NH3/H2 dataset

The private NH3/H2 dataset contains 162,404 sensor-response samples grouped into 378 gas exposures. Every measurement has 20 sensor-response channels, and the concentration range is 0–500 ppm for both gases. For the evaluation protocol, these exposures are divided into six non-overlapping partitions of 63 exposures. Each partition contains 57 unique (NH3,H2) concentration pairs and six repeated exposures.

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Partitions 1–4 are augmented for unlabeled pretraining. Five 64-sample windows are extracted from each exposure by shifting the starting offset from 0 to 4 time steps. Therefore, these four partitions contain 252 exposures and provide 1,260 pretraining windows. Partition 5 is used for labeled adaptation and contains 63 exposures, including 57 candidate concentration pairs. Finally, partition 6 contains the remaining 63 exposures and is reserved for testing without window augmentation. Only its offset-zero window is scored, so each test exposure contributes one prediction.

Complete exposures are assigned to a partition before window extraction, and scaling parameters are fitted using partitions 1–4 only. Partition 6 remains excluded from scaling, pretraining, concentration-pair selection, supervised fitting, and checkpoint selection to prevent data leakage.

### B. Dynamic Gas Mixture Dataset

The dynamic gas mixture dataset is publicly available from the UCI Machine Learning Repository [15], DOI 10.24432/C5WP4C. Its acquisition and continuous-monitoring protocol are described in [8]. This work uses the CO and ethylene portion, prepared as 245 chronological exposure runs with 16 response channels and 1,225 stored windows. Each run contributes five 64-sample windows at offsets 0, 20, 40, 60, and 80. The processed target scale is 0–533.33 ppm for CO and 0–20 ppm for ethylene.

The 245 runs are divided, in acquisition order, into five partitions of 49 runs. Partitions 1–3 provide 147 runs and 735 windows for unlabeled pretraining. Partition 4 provides 49 runs and 24 unique concentration pairs for labeled adaptation. Partition 5 provides 49 future runs for untouched evaluation. As in the NH3/H2 dataset, adaptation makes use of the data augmentation technique to create five windows from a selected exposure, whereas no data augmentation is applied for partition 5.

### C. Sensor Drift Dataset

To evaluate long-term drift, we use the public sensor drift dataset [16], DOI 10.24432/C5MK6M. The drift-compensation setting is described by Vergara et al. [17], and calibration with a limited number of experiments is examined in [14]. The dataset contains 13,910 complete records organized into ten time-based partitions spanning months 1–36. Each record holds 128 values: eight engineered descriptors for each of 16 sensor channels, so that each sample is a 16×8 matrix. The eight descriptors per sensor are two resistance-change features, three rising-response exponential-moving-average features, and three decaying-response exponential-moving-average features; they are engineered descriptors rather than consecutive temporal samples. The six gases are ethanol, ethylene, ammonia, acetaldehyde, acetone, and toluene.

Partitions 1–6, covering months 1–20, are used for self-supervised pretraining. Partition 7, collected in month 21, provides limited labeled adaptation data, and partitions 8–10, covering months 22–36, are retained as separate future tests. The adaptation budget is 20 selected records per gas; four records per gas are reserved for adaptation-only validation, leaving 16 per gas for supervised fitting in each repeat. Experiments use seeds 42, 142, and 242.

Raw temporal curves are unavailable in this benchmark, so the self-supervised masking is adapted to the representation. A complete sensor is masked independently with probability 0.18. One descriptor group—the resistance pair, the rising descriptors, or the decaying descriptors—is selected per record and hidden independently on each sensor with probability 0.35, and individual descriptors are additionally masked with probability 0.08. The shared principle remains structured reconstruction of physically grouped sensor information. The descriptor encoder maps the eight scaled descriptors and eight mask indicators through a shared 16→64→32 GELU multilayer perceptron, adds a learned 32-dimensional sensor-identity embedding, applies the same two-layer four-head Transformer, and pools mean and maximum across the 16 contextual tokens followed by a 64-unit GELU layer with dropout 0.15.

The supervised objective for this extension contains only the gas-conditional concentration term. Six gas-conditional regression heads output normalized log-concentration values, and the known gas identity selects the corresponding output; no gas-classification head is trained or evaluated. The reported concentration MAE is therefore an oracle-class conditional metric: it assumes gas identity is available and does not represent end-to-end concentration prediction for an unknown gas. Per-gas concentrations are mapped to normalized log space as in (17).

The split strategy for all three datasets is illustrated in Fig. 2. Pretrain indicates partitions for which no label was used. Adapt is the labeled partition and Test is untouched for evaluation.

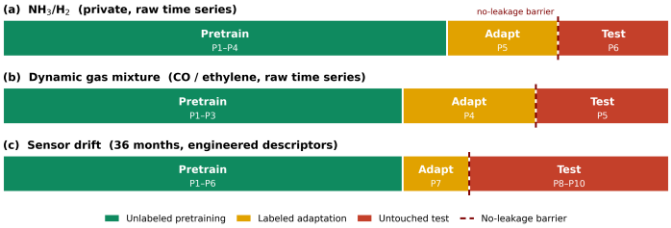



Fig. 2. Leakage-controlled evaluation across the three datasets.

### D. Preprocessing and Optimization

For the two raw-response datasets, a per-sensor robust scaler is fitted on unlabeled pretraining windows only. The center is the median and the scale is the interquartile range; scaled values are clipped to $[-12,12]$. Targets are divided by fixed implementation scales before Huber training: [500,500] ppm for the NH3/H2 dataset and [533.33,20] ppm for the dynamic gas mixture dataset. Predictions are transformed back to ppm and clipped to these ranges.

Raw-response pretraining uses AdamW for 60 epochs with learning rate $8 \times 10^{-4}$, weight decay $10^{-4}$, batch size 64, and gradient-norm clipping at 2. Supervised fitting uses 120 epochs and Huber loss with $\delta = 0.1$ in normalized target space. Scratch training uses learning rate $5 \times 10^{-4}$. Frozen transfer trains only the head at $10^{-3}$. Fine-tuning uses $8 \times 10^{-5}$ for the encoder and $8 \times 10^{-4}$ for the head. All supervised policies use weight decay $2 \times 10^{-4}$ and batches of at most 64 samples.

Sensor drift dataset features are robust-scaled using partitions 1–6 with the 10th–90th percentile range and clipped to $[-10,10]$. Descriptor pretraining uses 40 AdamW epochs, learning rate $8 \times 10^{-4}$, weight decay $10^{-4}$, batch size 256, and gradient clipping at 2. Supervised sensor drift dataset fitting uses at most 120 epochs, weight decay $2 \times 10^{-4}$, and

TABLE I. SUMMARY OF FULL-BUDGET REGRESSION PERFORMANCE

| Target | Scratch MAE | Best policy | Transfer MAE | Transfer RMSE | Transfer $R^2$ | Pooled $Q_{0.9}$ | Gain |
|---|---|---|---|---|---|---|---|
| NH3 | 18.47 ± 3.95 | Fine-tuned | **14.92 ± 0.57** | 19.83 ± 1.35 | 0.985 ± 0.002 | 33.62 | 19.2% |
| H2 | 36.60 ± 4.20 | Fine-tuned | **29.16 ± 1.10** | 41.05 ± 1.09 | 0.941 ± 0.003 | 71.90 | 20.3% |
| CO | 65.00 ± 9.96 | Frozen | **46.16 ± 4.58** | 73.28 ± 4.87 | 0.776 ± 0.029 | 118.82 | 29.0% |
| Ethylene | 1.992 ± 0.222 | Fine-tuned | **1.371 ± 0.025** | 2.196 ± 0.051 | 0.802 ± 0.009 | 3.819 | 31.2% |

adaptation-only early stopping with patience 20. The head learning rate is $8 \times 10^{-4}$; the encoder rate is $5 \times 10^{-4}$ from scratch and $8 \times 10^{-5}$ during fine-tuning. Concentration is transformed per gas as

$$u_i = \frac{\log(1 + c_i) - \mu_{r_i}}{\sigma_{r_i}}, \qquad (17)$$

where $r_i$ is the true gas class and $\mu_r, \sigma_r$ are estimated from the supervised fitting subset only. The sensor drift dataset Huber term uses $\delta = 0.5$.

### E. Evaluation Metrics

Besides using standard metrics like MAE, RMSE and $R^2$, we also measure the 90th percentile of absolute error , given by (18) to examine large errors.

$$Q_{0.9,j} = \text{quantile}_{0.9}\left(|y_{ij} - \hat{y}_{ij}|\right), \qquad (18)$$

Relative transfer gain is defined as in (19):

$$\text{Gain}(\%) = 100\,\frac{E_{\text{scratch}} - E_{\text{transfer}}}{E_{\text{scratch}}}, \qquad (19)$$

For sensor drift dataset concentration regression, we report the unweighted mean across six gases in (20),

$$\text{MacroMAE} = \frac{1}{6}\sum_{r=1}^{6} \text{MAE}_r\,. \qquad (20)$$

## V. RESULTS

### A. Self-Supervised Pretraining

The reconstruction objective converges on all three data settings as shown in Fig.3. For the NH3/H2 dataset, the reconstruction loss decreases from 2.1595 to 0.0762. For the dynamic gas mixture dataset, the loss reaches approximately 0.165 at epoch 51, while the final stochastic-mask loss is 0.3020. The difference is expected because a new random mask is generated over training and therefore the difficulty changes from epoch to epoch. The sensor drift dataset descriptor loss also falls steadily. Across three repeats, it moves from 0.1941–0.2118 initially to 0.0315–0.0321.

Reconstruction loss alone is not evidence of a good calibration model. A network could reconstruct smooth or redundant signals without preserving the information needed for concentration estimation. We therefore use the downstream experiments as the main test of representation quality.

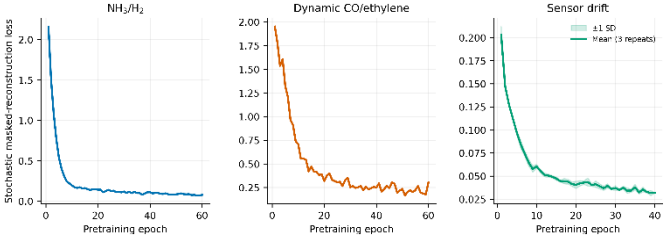



Fig.3. Self-supervised masked-reconstruction loss on all three data settings.

### B. Full-Budget

The overal result of the pipeline is shown in Table I. Compare to trainning from scratch, merging pretrain data with few known labels increase MAE significantly. As well as improved transfer MAE, the high $R^2$ and high gain indicate that the approach is effective in predicting the concentration of gases using very few labeled samples in different sensor settings. This pattern is repeated in three different runs with different seeds.

Table II reports scratch, frozen-transfer, and fine-tuned-transfer MAE separately for each target at the full label budget. Frozen and fine-tuned transfer are predefined policies; both are reported on the untouched test partition, and neither is chosen from test MAE. Transfer improves MAE for all four targets under both policies. Relative to scratch, fine-tuned transfer lowers NH3 MAE to 14.92 ppm (19.2%), H2 to 29.16 ppm (20.3%), and ethylene to 1.371 ppm (31.2%); for CO the frozen policy reaches 46.16 ppm (29.0%) while fine-tuning reaches 51.07 ppm (21.4%). Fig. 4 compares predicted and reference concentrations for the three policies on held-out exposures.

TABLE II. FULL-BUDGET REGRESSION MAE WITH DIFFERENT ADAPTATION POLICIES

| Target | Scratch MAE | Frozen-transfer MAE | Fine-tuned-transfer MAE |
|---|---|---|---|
| NH3 | 18.47 ppm | 17.57 ppm | 14.92 ppm |
| H2 | 36.60 ppm | 32.61 ppm | **29.16 ppm** |
| CO | **65.00 ppm** | **46.16 ppm** | 51.07 ppm |
| Ethylene | **1.992 ppm** | **1.503 ppm** | 1.371 ppm |

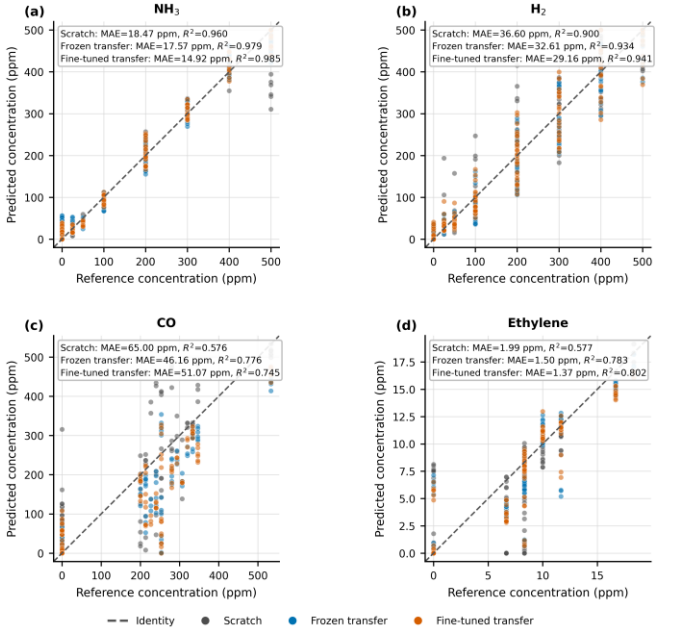



Fig. 4. Predicted versus reference concentration at the full adaptation budget for NH3, H2, CO, and ethylene. Scratch, frozen transfer, and fine-tuned transfer are shown together; points pool three repeats and the dashed line denotes ideal prediction.

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### C. Label-Efficient Calibration

The low-label experiments provide the clearest evidence for the intended use of pretraining. For CO, transfer reduces MAE by 47.7% with five labeled pairs and by 48.5% with ten pairs. H2 gains 33.0% at five pairs and 32.1% at ten pairs. Ethylene improves by 27.7% and 32.8% at the same two budgets.

The complete mean-MAE results used in the label-budget curves are shown in Table III, where pairs is the number of labeled adaptation pairs. Fine-tuned transfer is generally better than frozen transfer. The strongest low-budget gains occur for H2 and CO, whereas NH3 and ethylene vary less monotonically with the label budget. The pattern is intuitive. A scratch model must learn low-level temporal features, channel relations, and the concentration mapping from a very small labeled set. In contrast, the transferred model already contains a representation of the unlabeled sensor structure. The supervised stage can focus more directly on the final calibration mapping.

TABLE III. ALL BUDGETS REGRESSION PERFORMANCE

| Target | Pairs | Scratch | Frozen | Fine-tuned |
|---|---|---|---|---|
| NH3 | 5 | 52.87 | 47.85 | **46.16** |
| NH3 | 10 | 43.46 | 38.78 | **38.39** |
| NH3 | 20 | 31.33 | **26.61** | 27.54 |
| NH3 | 40 | 29.35 | 23.58 | **22.71** |
| NH3 | 57 | 18.47 | 17.57 | **14.92** |
| H2 | 5 | 143.82 | 102.36 | **96.43** |
| H2 | 10 | 103.62 | 76.42 | **70.36** |
| H2 | 20 | 53.79 | **40.93** | 41.23 |
| H2 | 40 | 46.42 | 39.52 | **38.62** |
| H2 | 57 | 36.60 | 32.61 | **29.16** |
| CO | 5 | 179.17 | 93.99 | **93.79** |
| CO | 10 | 115.06 | **59.27** | 68.42 |
| CO | 20 | 72.11 | **46.06** | 49.27 |
| CO | 24 | 65.00 | **46.16** | 51.07 |
| Ethylene | 5 | 3.255 | 2.407 | **2.352** |
| Ethylene | 10 | 3.422 | 2.391 | **2.301** |
| Ethylene | 20 | 1.934 | **1.424** | 1.507 |
| Ethylene | 24 | 1.992 | 1.503 | **1.371** |

### D. Multif-fold Validation

To check whether improvement depends on one favorable partition, the NH3/H2 dataset is rerun in six leave-one-partition-out folds. Each fold holds out one partition for testing, uses the preceding partition for adaptation (partition 6 when partition 1 is tested), and uses the other four partitions for unlabeled pretraining. Across 18 fold/repeat runs at 57 pairs, scratch obtains 33.01±17.81 ppm NH3 MAE and 48.78±19.27 ppm H2 MAE. Transfer obtains 27.60±10.37 and 38.97±10.13 ppm, respectively. Paired exposure-level improvement is 5.41 ppm for NH3, with a 95% bootstrap interval of 3.74–7.10 ppm, and 9.81 ppm for H2, with an interval of 7.56–11.94 ppm. Each gas contributes 1,134 paired predictions.

For multi-fold validation, the dynamic gas mixture dataset is divided into seven smaller chronological partitions and evaluated in three rolling pretrain–adapt–test folds. At the common 20-pair budget, scratch obtains 100.60±14.13 ppm CO MAE and 2.653±0.500 ppm ethylene MAE. Transfer obtains 80.07±41.83 and 2.267±0.989 ppm, respectively. CO improves by 20.53 ppm with a 95% interval of 10.51–30.45 ppm. Ethylene improves by 0.386 ppm with an interval of 0.121–0.661 ppm. Each gas contributes 315 paired predictions. All paired intervals remain above zero.

### E. Ablation Study

The ablation analysis shows that the architecture components do not contribute equally to all gases. Sensor identity is important for H2, CO, and ethylene. Temporal masking is particularly useful for NH3 and H2. Complete-sensor masking supports H2, CO, and ethylene. Cross-sensor attention improves three targets, but its effect is mixed for NH3.

No single component dominates every target. The full configuration is strongest on three targets, while NH3 favors a simpler attention ablation. Table IV shows MAE changes in ppm, averaged over three repeats. Positive values indicate higher error after component removal. The ablation protocol reserves 20% of adaptation groups for validation, so these differences are internal to that protocol.

TABLE IV. ABLATION STUDY PERFORMANCE

| Removed component | NH3 | H2 | CO | Ethylene |
|---|---|---|---|---|
| None: full model | +0.00 | +0.00 | +0.00 | +0.00 |
| Sensor-ID embedding | **+0.36** | **+8.86** | **+7.02** | **+0.25** |
| Cross-sensor attention | -1.52 | **+2.66** | **+8.28** | **+0.01** |
| Temporal-span masking | **+4.39** | **+5.01** | **+2.23** | +0.00 |
| Complete-sensor masking | -0.26 | **+1.44** | **+1.21** | **+0.23** |

### F. Baseline comparision

To compare the proposed transfer approach with traditional baseline models, we evaluate Extra Trees and XGBoost. In addition, the scratch model uses the same labeled subsets, architecture, training epochs, and untouched test data as the transfer models. Therefore, the comparison isolates the effect of unlabeled pretraining. Given the same training data and evaluation data, at full budget, the proposed approach gives the lowest MAE for all four targets as in Table V.

TABLE V. BASELINE MODELS COMPARISION MAE

| Dataset | Target | Scratch | Extra Trees | XGBoost | Transfer |
|---|---|---|---|---|---|
| NH3/H2 | NH3 | 18.47 | 17.70 | 20.07 | **14.92** |
| NH3/H2 | H2 | 36.60 | 33.00 | 31.97 | **29.16** |
| Dynamic gas mixture | CO | 65.00 | 76.58 | 46.44 | **46.16** |
| Dynamic gas mixture | Ethylene | 1.992 | 2.692 | 2.015 | **1.371** |

### G. Long-Term Sensor Drift: Concentration Regression

The sensor drift dataset contains six gases and spans 36 months. Partitions 1–6 are used for pretraining, partition 7 supplies limited labeled adaptation data, and partitions 8–10 are held out as future drift conditions. When the known gas identity selects the corresponding concentration head, transfer improves macro-MAE in every future partition, as shown in Table VI. In partition 8, scratch achieves MAE of 23.14 ppm, while fine-tuned transfer reaches 19.14 ppm, a 17.3% gain. In partition 9, fine-tuned transfer is best at 14.77 ppm compared with 21.06 ppm for scratch, a 29.9% gain. Partition 10 is much harder. Scratch MAE increases to 146.95 ppm. Frozen transfer reduces the error to 142.78 ppm, corresponding to a 2.8% gain, but the absolute error remains large. Thus, pretraining provides only a small improvement under the strongest drift condition.

TABLE VI. REGRESSION MAE ON SENSOR DRIFT DATASET

| Policy | Partition 8 *22nd and 23rd month* | Partition 9 *24th and 30th month* | Partition 10 *36th month* |
|---|---|---|---|
| Scratch | 23.14 ± 4.65 ppm | 21.06 ± 2.01 ppm | 146.95 ± 8.71 ppm |

| | | | |
|---|---|---|---|
| Frozen transfer | 20.98 ± 2.46 ppm | 20.48 ± 4.14 ppm | **142.78 ± 1.13 ppm** |
| Fine-tuned transfer | **19.14 ± 2.47 ppm** | **14.77 ± 2.92 ppm** | 142.97 ± 7.70 ppm |

### H. Tail Error and Predictive Uncertainty

Mean error can hide occasional large failures. We therefore also inspect the upper tail of absolute error. Transfer reduces large errors for NH3, H2, and CO. For ethylene, the tail behavior depends more strongly on the adaptation policy. This again supports the view that frozen and fine-tuned transfer are not interchangeable.

We also use split-conformal prediction as a diagnostic. Nominal 90% coverage is not achieved under the temporal shifts. Empirical coverage is 81.3% for NH3, 71.5% for H2, 59.0% for CO, and 63.8% for ethylene. These values are well below the nominal level, especially for the dynamic gas mixture dataset.

Although point predictions improve, the intervals remain poorly calibrated after distribution shift. They also become wider and less stable when the calibration set is small. Thus, some form of shift-aware calibration would be useful before deployment.

## VI. Discussion And Conclusion

The encoder appears to learn more than a generic compression of the input. Sensor identity helps with several targets, which suggests that the physical channel index carries useful information. Complete-sensor masking is also beneficial in most cases, indicating that a missing response can be partly inferred from the remaining array. Temporal-span masking plays a different role because it encourages the encoder to follow local response evolution instead of relying only on a final value. In the ablation study, this is particularly useful for NH3 and H2. Cross-sensor attention then combines the sensor tokens at array level, allowing the model to represent both within-sensor dynamics and relationships between sensors. At the same time, attention has a mixed effect for NH3, and no single component dominates every target. The observed gain therefore appears to come from the broader pretraining strategy rather than from one isolated layer.

The advantage of transfer is strongest when only a few labeled concentration pairs are available. With five or ten pairs, a scratch model must learn sensor features and the calibration map at the same time. The transferred model already contains structure learned from unlabeled measurements, so the supervised stage begins with a smaller learning problem.

The long-term drift experiment suggests that concentration-related structure learned from earlier partitions remains useful later in time. Regression may depend on response magnitude, descriptor trends, or correlations within each gas. Some of these relationships appear to persist even as the overall sensor response changes, although the present experiments do not identify which mechanism is most important. The large absolute error observed in the last drift partition also shows that pretraining alone is not sufficient. An explicit temporal- or domain-alignment objective added after sensor-aware pretraining could improve robustness under stronger drift.

The closest related studies already address quantitative gas prediction. Ni *et al.* [9] show that deep temporal-spatial feature fusion supports quantitative analysis, Tang *et al.* [10] demonstrate transfer learning for a compact sensor system, Xia *et al.* [11] focus on small-data mixed-gas monitoring, and OT-DARN [12] targets domain-adaptive concentration prediction. In our framework, pretraining requires no concentration labels and still preserves physical sensor identity. This makes it suitable for a sensing system that accumulates unlabeled measurements before a smaller controlled calibration campaign is performed. However, previous studies use different datasets and evaluation splits, so to the best of the authors' knowledge there is no suitable direct comparison between this work and previous studies.

In summary, this work examines whether unlabeled Electronic nose measurements can reduce the amount of labeled data needed for quantitative calibration. To do so, we introduced a sensor-aware self-supervised encoder with shared temporal feature extraction, physical sensor identity, cross-sensor attention, and structured masking. The underlying idea is straightforward: first learn recurring behavior in the sensor system, and then use labeled measurements to learn concentration.

Across the NH3/H2 dataset and dynamic gas mixture dataset, transfer improves concentration regression for all four targets. The largest benefits appear when only a small number of labeled calibration pairs are available. Moreover, multi-fold validation shows paired improvement across held-out NH3/H2 partitions and future chronological partitions.

On the 36-month sensor drift dataset, transfer improves oracle-class gas-conditional concentration MAE in every future partition. The improvement is smaller in partition 10, where the absolute error remains high, but the representation still retains useful quantitative structure under this protocol.

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*All code and datasets are available at: https://github.com/ngdvan/Transfer-learning-Enose*